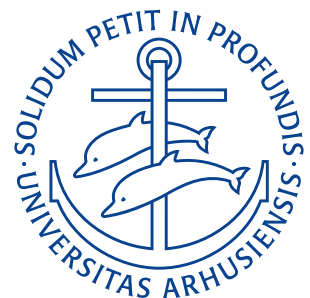


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Exercises

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Lecture 7: Euler and Bernoulli Equation

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Problem 6.11

An open-circuit wind tunnel draws in air from the atmosphere through a well-contoured nozzle. In the test section, where the flow is straight and nearly uniform, a static pressure tap is drilled into the tunnel wall. A manometer connected to the tap shows that static pressure within the tunnel is 45 mm of water below atmospheric. Assume that the air is incompressible, and at 25 °C, 100 kPa absolute. Calculate the air speed in the wind-tunnel test section.

Assumptions

- Steady, incompressible, inviscid, and uniform flow
- The air behaves as an ideal gas

Derivation

The Bernoulli equation states that

$$\frac{p}{\rho} + \frac{V^2}{2} + gz = \text{constant}.$$

In this case we assume only horizontal flow, hence $gz = 0$, furthermore, the air is assumed to be stationary at the entrance, hence $V_1 = 0$. Therefore

$$\frac{p_1}{\rho} = \frac{p_2}{\rho} + \frac{V_2^2}{2} \implies V_2 = \sqrt{2 \cdot \frac{p_1 - p_2}{\rho}}.$$

The manometer reading corresponds to

$$p_1 - p_2 = \rho_{\text{H}_2\text{O}} g h.$$

Hence,

$$V_1 = \sqrt{\frac{2 \rho_{\text{H}_2\text{O}} g h}{\rho}}.$$

The density of the air can be found using the ideal gas law

$$\rho = \frac{p}{RT} = \frac{100 \text{ kPa}}{287 \frac{\text{J}}{\text{kgK}} \cdot 298 \text{ K}} = 1,17 \frac{\text{kg}}{\text{m}^3}.$$

And therefore the speed in the wind-tunnel can be calculated as

$$V_1 = \sqrt{\frac{2 \cdot 1000 \frac{\text{kg}}{\text{m}^3} \cdot 9,81 \frac{\text{m}}{\text{s}^2} \cdot 45 \text{ mm}}{1,17 \frac{\text{kg}}{\text{m}^3}}} = 27,5 \frac{\text{m}}{\text{s}}.$$

$V_1 = 27,5 \frac{\text{m}}{\text{s}}$

RESULT

Problem 6.9

In a pipe 0,3 m in diameter, $0,3 \frac{\text{m}^3}{\text{s}}$ of water are pumped up a hill. On the hilltop (elevation 48), the line reduces to 0,2 m diameter. If the pump maintains a pressure of 690 kPa at elevation 21, calculate the pressure in the pipe on the hilltop.

Assumptions

- Elevation 48 and 21 correspond to $z = 48 \text{ m}$ and $z = 21 \text{ m}$ respectively
- Steady, incompressible, inviscid, and uniform flow

Derivation

From continuity the flow rate of water must be equal at all sections of the pipe. Therefore we can employ the continuity equation as

$$Q = A_1 V_1 = A_2 V_2.$$

Hence the flow speeds at both elevations can be found as

$$V_1 = \frac{Q}{A_1} = \frac{0,3 \frac{\text{m}^3}{\text{s}}}{\pi \cdot \frac{(0,3 \text{ m})^2}{4}} = 4,24 \frac{\text{m}}{\text{s}}$$

$$V_2 = \frac{Q}{A_2} = \frac{0,3 \frac{\text{m}^3}{\text{s}}}{\pi \cdot \frac{(0,2 \text{ m})^2}{4}} = 9,55 \frac{\text{m}}{\text{s}}.$$

Now we can use the Bernoulli equation

$$\begin{aligned} \frac{p_2}{\rho} + \frac{V_2^2}{2} + g z_2 &= \frac{p_1}{\rho} + \frac{V_1^2}{2} + g z_1 \\ p_2 &= p_1 + \rho \frac{V_1^2 - V_2^2}{2} + \rho g (z_1 - z_2) \\ &= 690 \text{ kPa} + 1000 \frac{\text{kg}}{\text{m}^3} \cdot \frac{(4,24 \frac{\text{m}}{\text{s}})^2 - (9,55 \frac{\text{m}}{\text{s}})^2}{2} + 1000 \frac{\text{kg}}{\text{m}^3} \cdot 9,81 \frac{\text{m}}{\text{s}^2} \cdot (21 \text{ m} - 48 \text{ m}) \\ &= 389 \text{ kPa}. \end{aligned}$$

$p_2 = 389 \text{ kPa}$

RESULT

Problem 6.2

The velocity field for a two-dimensional downward flow of water against a plate is given by $\mathbf{V} = Ax\hat{i} - Ay\hat{j}$. Plot the pressure gradient along the centerline and determine the pressure gradient at $(x, y) = (0, 2)$, $(0, 0)$, and $(2, 0)$.

Assumptions

- Frictionless and incompressible flow

Derivation

To find the pressure gradient we use Euler's equation

$$\rho (\mathbf{V} \cdot \nabla) \mathbf{V} = \rho \mathbf{g} - \nabla p.$$

The particle acceleration is defined as

$$\mathbf{a}_p = (\mathbf{V} \cdot \nabla) \mathbf{V}$$

which on scalar form is

$$\begin{aligned} a_{x_p} &= u \frac{\partial u}{\partial x} + v \frac{\partial u}{\partial y} + \frac{\partial u}{\partial t} \\ a_{y_p} &= u \frac{\partial v}{\partial x} + v \frac{\partial v}{\partial y} + \frac{\partial v}{\partial t}. \end{aligned}$$

Thus

$$\begin{aligned} a_{x_p} &= A^2 x \\ a_{y_p} &= A^2 y. \end{aligned}$$

Hence, the total acceleration is

$$\mathbf{a}_p = A^2 x + A^2 y.$$

Now we can solve the Euler equation for the pressure gradient

$$\nabla p = -\rho \mathbf{g} - \rho (A^2 x + A^2 y).$$

As gravity is in the y -direction this can also be written as

$$\nabla p = -\rho A^2 x \hat{i} - \rho (g + A^2 y) \hat{j}.$$

The pressure gradient at the three points is therefore

$$\begin{aligned} \nabla p(0, 2) &= -\rho (g + 2A^2) \hat{j} \\ \nabla p(0, 0) &= -\rho g \hat{j} \\ \nabla p(2, 0) &= -2\rho A^2 \hat{i}. \end{aligned}$$

RESULT

Problem 6.18

Consider frictionless, incompressible flow of air over the wing of an airplane flying at $200 \frac{\text{km}}{\text{h}}$. The air approaching the wing is at 65 kPa and -10°C . At a certain point in the flow, the pressure is 60 kPa. Calculate the speed of the air relative to the wing at this point and the absolute air speed.

Assumptions

- Incompressible, inviscid, steady, and horizontal flow along a streamline

Derivation

We start by finding the density of the air at the given conditions

$$\rho = \frac{p}{RT} = \frac{65 \text{ kPa}}{287 \frac{\text{J}}{\text{kgK}} \cdot 263 \text{ K}} = 0,861 \frac{\text{kg}}{\text{m}^3}.$$

Now we can apply the Bernoulli equation as

$$\begin{aligned} \frac{p_1}{\rho} + \frac{V_1^2}{2} &= \frac{p_2}{\rho} + \frac{V_2^2}{2} \\ V_2 &= \sqrt{V_1^2 + 2 \cdot \frac{p_1 - p_2}{\rho}} \\ &= \sqrt{\left(200 \frac{\text{km}}{\text{h}}\right)^2 + 2 \cdot \frac{65 \text{ kPa} - 60 \text{ kPa}}{0,861 \frac{\text{kg}}{\text{m}^3}}} \\ &= 121 \frac{\text{m}}{\text{s}}. \end{aligned}$$

This corresponds to an absolute velocity of

$$V_{2\text{abs}} = V_2 - V_1 = 65,4 \frac{\text{m}}{\text{s}}$$

RESULT